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Technical Paper

Processing indoor positioning data by goal-oriented supervised fuzzy clustering for tool management

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ABSTRACT

Indoor positioning systems allow real-time tracking of tool locations. Tool utilization can be calculated based on positional data of the storage and manufacturing areas. Due to the uncertainty of the position measurements, estimation of the state of the tools is problematic when the distance between the examined zones is less than the estimation error. We propose a goal-oriented supervised fuzzy clustering algorithm that utilizes the activity state of the tool, as the algorithm simultaneously maximizes the spatial distribution probability and the probability of a specific activity state occurring in a cluster. By weighting data points according to the time spent in the related states and positions, the resulting cluster weights can be interpreted as tool utilizations. The applicability of the developed method is presented through the processing of position data from crimping tools used by a wire harness manufacturer.

1. Introduction

Modern and flexible manufacturing systems (FMSs) utilize multi-functional machines with the help of various replaceable tools. As a result, the number of tools in a medium-sized factory can exceed the order of a thousand [1]. Tooling costs can be responsible for 25–30% of the overall manufacturing costs [2]. Tooling cost can be reduced by the careful selection of the number of tools [3], by the optimization of the manufacturing process [4,5], and by the application of tool management systems. Due to the importance and complexity of these problems, tool management is an essential and long-studied area of production management [6].

In the future, tool management should be integrated into Industry 4.0 framework. Industry 4.0-based tool management should not only utilize quality and production management methods, such as Six Sigma, Lean, Lean Six Sigma, the Theory of Constraints, and Total Quality Management. The need for sustainability has led to the Zero-Defect Manufacturing (ZDM) concept [7] that aims to improve quality in manufacturing systems with a goal-oriented prediction-prevention strategy [8].

Effective tool management requires computer support [9] and information integration [10]. Tool management requires important information on the tool location and condition and utilization of tools as well as the production schedule, necessitating an integrated database and tool tracking system [11]. Perera et al. [12] highlighted that the companies which do not have sufficient data about their tooling are unaware of tooling problems, so there is a significant loss if the necessary information is not available.

This work aims to present an algorithm that enables the utilization of the uncertain measurements of indoor positioning systems (IPS) in tool management.

When tool positional information is not available, workers have to search the tools for over 20% of their working hours [11] and production is stopped for nearly 15% of the time due to the unavailability of tools [2]. When the tools circulating in the factory cannot be found, the workers increase the size of the local inventory to enhance availability at their workstations. In such cases, the tool concerned does not return to its storage location but stays at the machine remains poorly utilized, which is responsible for 15% of the tool inventory [11]. Furthermore, changeover times and downtimes can also significantly increase [13]. Therefore, in some companies, the amount of time spent finding the right tool can constitute 30% of the total preparation time of the tool [11].

IPSs can generate valuable tracking data about the position of tools

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[14]. In the case of flexible production systems, the application of an IPS is crucial for production monitoring [15]. Real-time tracking data can be converted into information that helps to control cycle times [16], explore the value streams [17] and integrate digital twins [18].

The data-based identification of the production zones and the calculation of tool utilization are not straightforward tasks. The first problem is related to data uncertainty. In workshops and production halls, the layout can constantly change so that tracked tools can be obscured by high shelves, machines, and metal surfaces. Consequently, some non-line-of-sight (NLOS) places often have no direct view from the IPS anchor points. As a result, position data are scattered depending on the accuracy of the IPS at specific locations. This measurement noise introduces uncertainty inherent to these systems, so IPS applications often require advanced data-processing solutions, like Kalman filters [19] or clustering algorithms [20]. Fuzzy logic can also handle measurement imprecision of localization [21]. Since imprecision is connected mainly to measurement errors, the probabilistic approach of the problem is also beneficial [22].

In this work, we present a method that integrates the benefits of fuzzy logic and probabilistic approach with the help of probabilistic Gaussian mixture models that can be interpreted as fuzzy rules [23]. The key idea is that the Gaussian mixture model can approximate the spatial distribution of the tools, so the model can be used to segment the shop floor into zones. When information about the activity of tools is also available, we can use this during the clustering resulting in conditional cluster probabilities that can be interpreted as tool utilizations.

The novelty of this work is a new clustering algorithm that is suitable to convert noisy tracking data into business value by segmentation of the shop floor and calculation of the tool utilization at the identified locations.

The structure of the paper and the main contributions of the work are as follows. We propose a supervised fuzzy clustering-based tool monitoring method in Section 2. After problem formulation is presented in Section 2.1, the determination of tool activity is discussed in Section 2.2, which will be followed by the mathematical formalization of the fuzzy clustering problem in Section 2.3. The methodological introduction ends with the discussion of the practical applicability of the results concerning clustering in Section 2.4 and finally with the definition of the number of clusters in Section 2.5. The effectiveness of the proposed methodology is presented in terms of the analysis of the position data of crimping tools used in cable manufacturing machines in Section 3. Section 4 presents the potential applications of the method. Finally, some concluding remarks are made.

2. The supervised fuzzy clustering-based tool monitoring method

The identification of relevant operations or operating zones from noisy position data can be interpreted as finding the distinct distributions generating a mixture of data distributions. The positioning system allows activities of tools that can be used during this identification process to be determined.

2.1. Problem formulation

By turning this problem into a data-related task, we propose a supervised fuzzy clustering (SFC)-based method for the detection of distinct clusters of manufacturing position data. The methodology is summarised in the following points:

1. Feature transformation: determination of tool activity.
2. SFC-based modelling of tool positions.
3. Zone categorization and utilization calculation.
4. Validation of the number of clusters.

The problem is formalized as follows. It is assumed that with the help of an internal positioning system, at time t, the position estimation of tool l ($l = 1, \ldots, L$) is recorded. The timestamp of the *nth* record on tool l is denoted by t_n^l. The position of tool l at time t_n^l is stored in the vector $\mathbf{x}_n^l$. In our example, the position vector consists of the raw horizontal ($x_{1,n}$) and vertical ($x_{2,n}$) coordinates of the located position $\mathbf{x}_n^l = \left[x_{1,n}^l, x_{2,n}^l\right]$. The set of the N_l observations of the lth tool is stored in a matrix of the observation data $\mathbf{X}^l = \left[\mathbf{x}_1^l \ldots, \mathbf{x}_{N_l}^l\right]$. The matrix containing the observation data of all tools is denoted as $\mathbf{X} = \left[[\mathbf{X}^1]^T, \ldots, [\mathbf{X}^L]^T\right]^T$.

However, as the recorded position information is obtained based on the wireless communication of the receiver unit and a small tag fixed on the monitored tool, the recorded position data are somewhat inaccurate, the inaccuracy of which is a function of its actual position (mainly due to radio frequency interference). This inaccuracy, paired with the stochastic working habits of the human workforce (moving the tools, placing them at arbitrary places based on the convenience of the workers, taking breaks, etc.), makes the monitoring of tool usage an extremely complex task. Due to these problems the usage of a tool cannot be measured simply according to its position or movement, as neither of these pieces of information clearly determines the working conditions. However, besides the recorded position, the activity state of the monitored tool can be derived from the sampling time and the speed at which the tool moves as will be discussed in the following section. The activity state of tool l during the *nth* observation is marked as $y_n^l \in \{r_1, \ldots, r_R\}$ and describes whether the tool is stationary, being moved or actively in use at the time.

Due to the inaccurate nature of the information, instead of the simple analysis of the raw position data to monitor tool usage, an advanced clustering algorithm is proposed to obtain accurate information on tool activities. Our task is to partition the data in matrix $\mathbf{Z}^l = [\mathbf{z}_1^l, \ldots, \mathbf{z}_n^l, \ldots, \mathbf{z}_N^l]$ into C clusters. The data matrix $\mathbf{Z}^l$ consists of $\mathbf{z}_n^l = \left[\mathbf{x}_n^{l\,T}, y_n^l\right]$ row vectors, which include the input $\mathbf{x}_n^l$ and output data (activity class labels) y_n^l of the *nth* observation on the *lth* tool.

By applying the proposed SFC approach, the time-weighted activity rate of a specific zone in the production directly represents the utilization of the applied tool in the related zone. As a result, the active and inactive zones are identified simultaneously according to the activity rate of the tool in the related zone, moreover, a more accurate evaluation of tool usage is made. For the sake of simplicity, the feature transformation and clustering algorithms are given in the case of one tool ($\mathbf{X} = \mathbf{X}^l$),therefore, the l superscript is omitted in the following subsections.

2.2. Feature transformation – the determination of tool activity

Even though an IPS primarily records the position of the monitored tag, further information can be derived from the recorded data.

The Δt sampling time, namely the time difference between two consecutive position measurements on the same tag, is a function of the activity of the tag due to energy-saving considerations: each tag is equipped with a gyroscope sensor, which indicates whether the tool is actively moved or not. If the tag is moved (or subject to vibrations), its sampling frequency is much higher, 2–10 s, while if the tag is inactive, its sampling frequency is reduced to $\sim$ 30 s. The time difference between two consecutive measurements is simply formulated as $\Delta t_n = t_n - t_{n-1}$. Similarly, the Euclidean distance between two consecutive position points can be calculated as well: $\Delta x_n = D(\mathbf{x}_n, \mathbf{x}_{n-1})$. As we obtained the time and distance between two consecutive data points on the same tool, naturally the speed of the tag can be subsequently calculated as follows: $v_n = \frac{\Delta x_n}{\Delta t_n}$.

Based on the available and derived variables, R number of different states are assigned to the monitored tool. In the present work, utilising the time difference between the recorded positions before (Δt_{n-1}) and

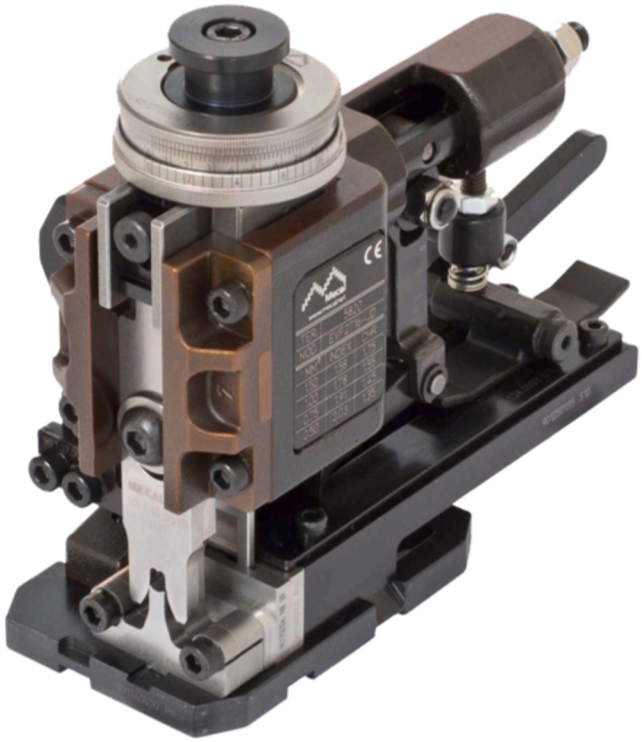


Fig. 1. A crimping tool used in cable production.

after (Δt_n) the data point on the same tag, the arriving (v_{n-1}) and leaving (v_n) speed of the tag at the recorded position as well as the corresponding user-defined thresholds (Δt_{thr} and v_{thr} for the sampling time and the speed of the tag, respectively), three states ($R = 3$) are defined, symbolizing the activity of the monitored tool:

- The tool is stationary and is not being actively used ($y_n = r_1 | \Delta t_{n-1} > \Delta t_{thr}$, $\Delta t_n > \Delta t_{thr}$).
- The tool is stationary, but is actively being used ($y_n = r_2 | \Delta t_{n-1} < \Delta t_{thr}$, $\Delta t_n < \Delta t_{thr}$, $v_{n-1} \leq v_{thr}$, $v_n \leq v_{thr}$).
- The tool is moving (or position data of the tag is noisy, showing quick jumps over long distances) ($y_n = r_3 | v_{n-1} > \Delta v_{thr}$, $v_n > \Delta v_{thr}$).

From the aforementioned list of activity states, only the ones where the tool is stationary or being used are of primary interest, therefore, in our analysis, the first two states, r_1 and r_2, are considered.

2.3. Supervised fuzzy clustering for tool monitoring

The methodology is based on the idea that the spatial distribution of position data originating from different sources (with optimally different statistical characteristics) represents the probability of finding a given tool at a given position in a given state represented by a specific class label. Our goal is to determine the distribution of the data. For this purpose, a clustering algorithm is applied. The number of clusters is denoted by C, which describes the spatial distribution of the data considering the R number of states.

The probability distribution of finding that particular tool at a given position in a given state ($p(\mathbf{x}_n, y_n)$) is approximated by elementary Gaussian distributions. Furthermore, the joint distribution of these local distributions can be broken down further into conditional probabilities:

$$p(\mathbf{x}_n, y_n) = \sum_{i=1}^{C} p(\mathbf{x}_n, y_n, c_i) = \sum_{i=1}^{C} p(\mathbf{x}_n, y_n | c_i) p(c_i) = \\ = \sum_{i=1}^{C} p(y_n | \mathbf{x}_n, c_i) p(\mathbf{x}_n | c_i) p(c_i) = \sum_{i=1}^{C} p(y_n | c_i) p(\mathbf{x}_n | c_i) p(c_i) \tag{1}$$

where c_i ($i = 1 \ldots C$) denotes the ith distribution component or cluster and $p(c_i)$ stands for the prior probability of distribution component c_i, representing the probability that an arbitrary data point is generated by the distribution c_i, which is calculated as the ratio of the data points in this cluster to the whole data set. Since the probability of finding a data point with a specific class label depends on the local distribution that generated it rather than its exact position: $p(y_n | \mathbf{x}_n, c_i) = p(y_n | c_i)$.

The $p(\mathbf{x}_n | c_i)$ conditional probability models the density of data point $\mathbf{x}_n$ belonging to distribution component c_i. It can be expressed as:

$$p(\mathbf{x}_n | c_i) = \left(\frac{1}{|2\pi \mathbf{F}_i|^{d/2}} \left(-\frac{1}{2} (\mathbf{x}_n - \mathbf{v}_i)^T (\mathbf{F}_i)^{-1} (\mathbf{x}_n - \mathbf{v}_i) \right) \right) \tag{2}$$

where d denotes the number of dimensions, $\mathbf{v}_i$ (Eq. (4)) stands for the cluster center, which is a d element vector, and $\mathbf{F}_i$ (Eq. (5)) represents the covariance matrix of the distribution, which is a matrix with dimensions $d \times d$.

The parameters of the joint distribution model of Eq. (1) can be obtained by the Expectation-Maximization (EM) algorithm. First, the model is initialized by assigning random values to the parameters. In the *Expectation* step, assuming the values these parameter are correct, the $p(c_i | \mathbf{x}_n, y_n)$ posterior probability is computed, which can be expressed by Bayes rule:

$$p(c_i | \mathbf{x}_n, y_n) = \frac{p(\mathbf{x}_n, y_n | c_i) p(c_i)}{p(\mathbf{x}_n, y_n)} = \frac{p(\mathbf{x}_n, y_n | c_i) p(c_i)}{\sum_{i=1}^{C} p(\mathbf{x}_n, y_n | c_i) p(c_i)} \tag{3}$$

In the *Maximization* step, the current distribution model is assumed to be correct and the parameters that maximize its likelihood are calculated as follows:

$$\mathbf{v}_i = \frac{\sum_{n=1}^{N} p(c_i | \mathbf{x}_n, y_n) \mathbf{x}_n}{\sum_{n=1}^{N} p(c_i | \mathbf{x}_n, y_n)} \tag{4}$$

$$\mathbf{F}_i = \frac{\sum_{n=1}^{N} p(c_i | \mathbf{x}_n, y_n) (\mathbf{x}_n - \mathbf{v}_i)(\mathbf{x}_n - \mathbf{v}_i)^T}{\sum_{n=1}^{N} p(c_i | \mathbf{x}_n, y_n)} \tag{5}$$

$$p(c_i) = \frac{1}{N} \sum_{n=1}^{N} p(c_i | \mathbf{x}_n, y_n) \tag{6}$$

$$p(y_n | c_i) = \frac{\sum_{n | y_n = r_i} p(c_i | \mathbf{x}_n, y_n)}{\sum_{n=1}^{N} p(c_i | \mathbf{x}_n, y_n)} \tag{7}$$

The expectation and maximization steps iterate until the parameters converge to a certain value.

This alternating optimization can be interpreted as an SFC task [23]. The fuzzy partition represented by $\mathbf{U} = [\mu_{i,n}]_{C \times N}$ matrix, where the degree of membership ($\mu_{i,n}$) represents, how the $\mathbf{z}_n$ measurement is in c_i cluster. The degree of membership can be expressed as the posterior probability $p(c_i | \mathbf{x}_n)$ of the cluster given the data point. The clustering algorithm minimizes the sum of weighted $D^2_{i,n}$ squared distances between the data points and the ith cluster centroid represented by η_i.

$$J(\mathbf{Z}, \mathbf{U}, \eta) = \sum_{i=1}^{C} \sum_{n=1}^{N} (\mu_{i,n})^m D^2_{i,n}(\mathbf{z}_n, c_i) \tag{8}$$

Where m denotes the fuzzy weighting exponent determining the fuzziness of the resulting clusters: $m = 1$ means a hard partition of the data points and the cluster centroid is the mean of the cluster members. As m increases, the partition becomes fuzzy and the cluster means are equal to the grand mean of $\mathbf{Z}$. Conventionally, $m = 2$. The cost function is subject to the restrictions on the $\mu_{i,n}$ membership values:

$$\mu_{i,n} \in [0, 1] \forall i, n; \\ \sum_{i=1}^{C} \mu_{i,n} = 1 \forall n; \\ 0 < \sum_{n=1}^{N} \mu_{i,n} < N \forall i \tag{9}$$

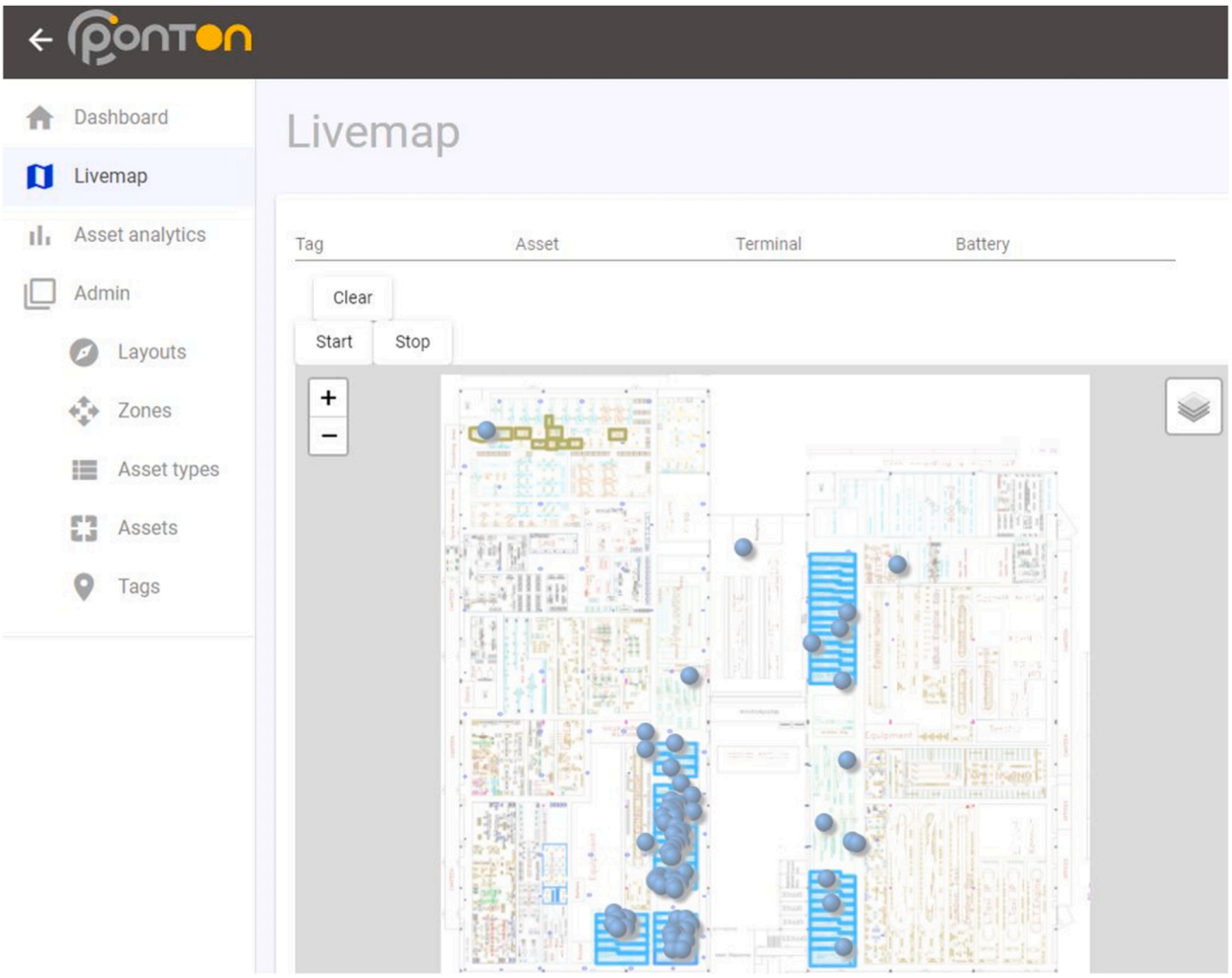


Fig. 2. The live map of the positioning system.

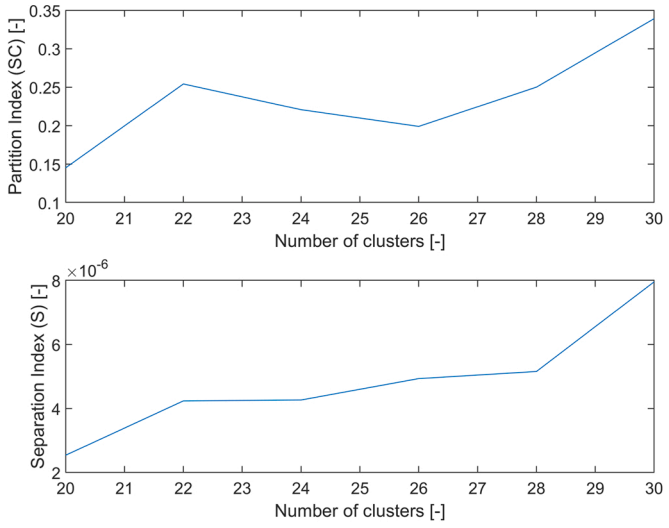


Fig. 3. Determination of the number of clusters. The upper plot shows the partition index values as a function of the number of clusters and the curve has a local maximum at 22. The lower plot shows the separation index as a function of number of clusters, where the curve starts to flatten out at 22. Based on these plots, 22 seems to be the optimal number of clusters.

The membership values can be interpreted as probabilities: $\mu_{i,n} \cong p(c_i|\mathbf{x}_n, y_n)$. The positional data is not equally sampled in time, so the $\mu_{i,n}$ membership degrees should be weighted by Δt_n time to represent temporal probability.

The distance is based on the joint probability of finding a data point in a given position with a given class label (Eq. (1)):

$$\frac{1}{D^2_{i,n}(\mathbf{z}_n, c_i)} = p(y_n|c_i)p(\mathbf{x}_n|c_i)p(c_i) = \left(-\frac{1}{2}(\mathbf{x}_n - \mathbf{v}_i)^T(\mathbf{F}_i)^{-1}(\mathbf{x}_n - \mathbf{v}_i)\right)p(y_n|c_i) = \frac{p(c_i)}{|2\pi\mathbf{F}_i|^{d/2}} \tag{10}$$

The alternating optimization algorithm includes the following steps:

Initialization: Given a dataset **Z**, determine the C number of clusters, the termination tolerance $\epsilon > 0$ and the weighting exponent m (which is equal to 2 in our case). Initialize the $\mathbf{U} = [\mu_{i,n}]_{C\times N}$.

Repeat for $k = 1, 2\ldots$ (k is the iteration counter).

Step 1 Parameter estimation

- Calculate the mean vectors, covariance matrices and prior probability of the Gaussian components.

$$v_i^{(k)} = \frac{\sum_{n=1}^{N}\left(\mu_{i,n}^{(k-1)}\right)^m \mathbf{x}_n \Delta t_n}{\sum_{n=1}^{N}\left(\mu_{i,n}^{(k-1)}\right)^m \Delta t_n} \tag{11}$$

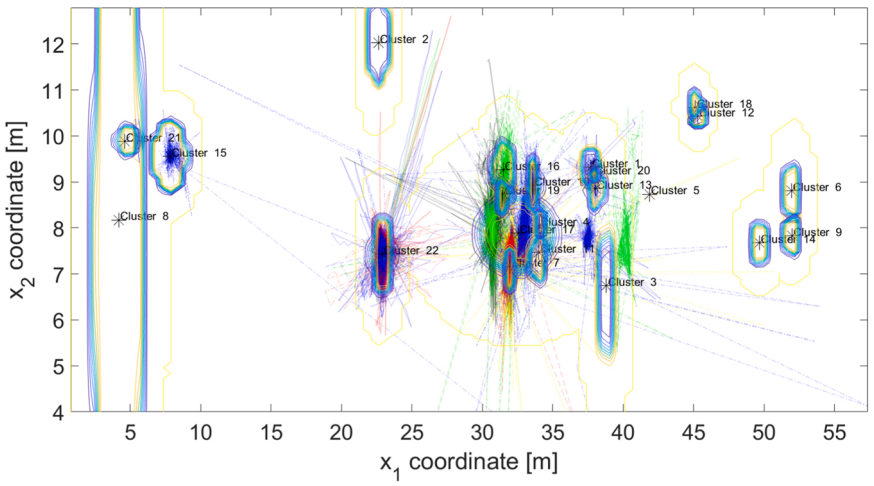


Fig. 4. The results of the SFC of tool position data: the cluster centers are indicated by black stars, while the zones with equal probability of tools being located there are represented by the contour plots. The position data of different tools are denoted by lines in different colors and styles. Some clusters are so close to each other that they overlap, which may be caused by these clusters representing different locations on the same desktop storage.

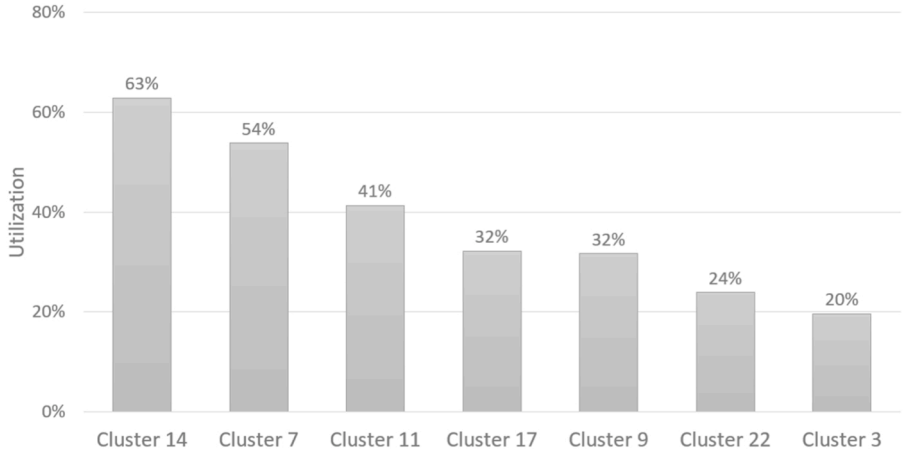


Fig. 5. The utilization values of the machines in percentage. The columns illustrate the temporal probability of finding active data points in a cluster represents machine.

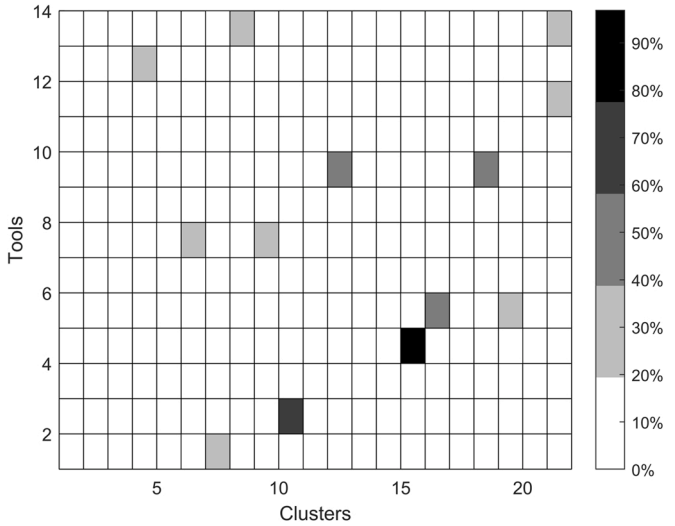


Fig. 6. The heat map of the utilization values of tools on the different machines. The darker the color, the higher the utilization of the tool.

$$\mathbf{F}_i^{(k)} = \frac{\sum_{n=1}^{N} \left(\mu_{i,n}^{(k-1)}\right)^m (\mathbf{x}_n - \mathbf{v}_i)(\mathbf{x}_n - \mathbf{v}_i)^T \Delta t_n}{\sum_{n=1}^{N} \left(\mu_{i,n}^{(k-1)}\right)^m \Delta t_n} \tag{12}$$

$$p(c_i)^{(k)} = \frac{\sum_{n=1}^{N} \left(\mu_{i,n}^{(k-1)}\right)^m \Delta t_n}{\sum_{n=1}^{N} \Delta t_n} \tag{13}$$

- Calculate the consequent probability parameters.

$$p(y_n = r_l | c_i)^{(k)} = \frac{\sum_{n|y_n=r_l} \left(\mu_{i,n}^{(k-1)}\right)^m \Delta t_n}{\sum_{n=1}^{N} \left(\mu_{i,n}^{(k-1)}\right)^m \Delta t_n} \tag{14}$$

Step 2 Compute the distances according to Eq. (10).

Step 3 Update the partitioned matrix

$$\mu_{i,n}^{(k)} = \frac{1}{\sum_{j=1}^{C} \left(D_{i,n}(\mathbf{z}_n, c_i)\right) / \left(D_{j,n}(\mathbf{z}_n, c_j)\right)^2 / (m-1)} \tag{15}$$

Until $||\mathbf{U}^{(k)} - \mathbf{U}^{(k-1)}|| < \epsilon$.

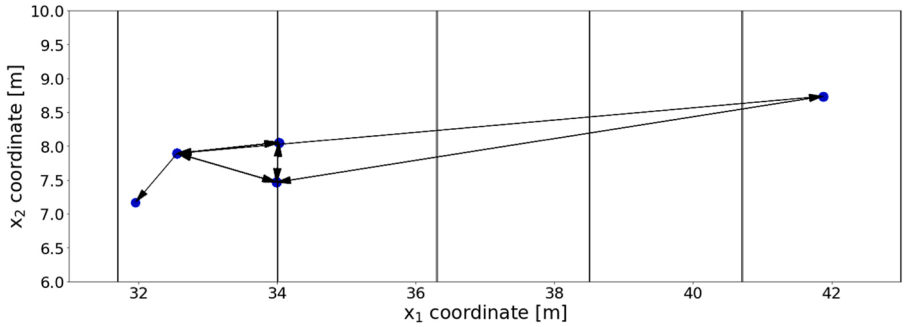


Fig. 7. The path of tool with ID 12. The blue circles represent the different cluster centers where the tool has been.

2.4. Zone categorization and calculation of tool utilization

The $p(y_n = r_2|c_i)$ conditional probability carries important information. Based on the probability of finding a data point in an active state in a given environment represented by c_i cluster, this environment can be categorized according to its function. For example, if a specific place represents a machine, this probability will be higher. In contrast, the probability of a storage-type cluster finding data points with active class label (while $p(y_n = r_1|c_i)$ takes a higher value) will be low. For the purpose of categorization, a decision threshold value α should be determined. If the conditional probability of finding an active data point in the given cluster is higher than this value, this cluster can be considered to be a machine. If not, it should be regarded as a storage area. If $p(y_n = r_2|c_i) > \alpha \Rightarrow c_i$ is considered to be machine.

In the case of clustering the position data of all the tools together, this probability will represent the utilization of the tools available in the vicinity of that machine which can be interpreted as the utilization of that machine:

$$U_i = p(y_n = r_2|c_i) \tag{16}$$

It should be noted that cases can occur when inactive and active tools are in the machine simultaneously (due to operators having left a tool there following the previous operation), which decreases the value of this probability. It can be said that when there is an active tool in the vicinity of the machine over a given period, then only that tool is counted. This method allows the utilization of a machine to be calculated but this is not our goal. We aim to calculate the utilization of specific tools in specific regions. The utilization of a single tool can be calculated as the ratio of its working hours to the whole period ($T = \sum_{n=1}^{N_l} \Delta t_n^l$). The working hours are known as the summation of the time differences related to data points with active class label:

$$U^l = \frac{\sum_{n=1}^{N_l} I({y_n}^l = r_2)\Delta {t_n}^l}{T} \tag{17}$$

The utilization value of tool l can be determined for machine c_i in two ways. On the one hand, it can be calculated as the ratio of the summation of time-weighted posterior probabilities of c_i to the whole time period, but on the other hand, if we build the model from the data of one single tool, $p(y_n = r_2|c_i)$ can be interpreted as the utilization of tool l on machine c_i:

$$U_i^l = \frac{\sum_{n=1}^{N_l} p(c_i|\mathbf{x}_n^l, y_n^l)\Delta t_n^l}{T} \cong p(y_n = r_2|c_i) \tag{18}$$

By totaling this for all clusters, the utilization of that particular tool can be determined:

$$U^l \cong \sum_{i=1}^{C} U_i^l \tag{19}$$

2.5. Definition of the number of clusters

The application of the algorithm requires the careful selection of the number of clusters. The optimal number of clusters can be evaluated by running the algorithm multiple times with different numbers of clusters and the monitoring cluster validity measures, like partition index (SC) and the separation index (S) [24]. The partition index is the ratio of the sum of the compactness of the clusters to their separation. It is expressed by:

$$SC = \sum_{i=1}^{C} \frac{\sum_{n=1}^{N_l} (\mu_{i,n})^m \left|\left|\mathbf{x}_n - \mathbf{v}_i\right|\right|}{\sum_{n=1}^{N_l} \mu_{i,n} \sum_{j=1}^{C} \left|\left|\mathbf{v}_j - \mathbf{v}_i\right|\right|} \tag{20}$$

While the partition index uses the fuzzy cardinality as the separation of clusters, the separation index uses the minimum distance:

$$S = \frac{\sum_{i=1}^{C} \sum_{n=1}^{N_l} (\mu_{i,n})^2 \left|\left|\mathbf{x}_n - \mathbf{v}_i\right|\right|}{N \min_{i,j} \left|\left|\mathbf{v}_j - \mathbf{v}_i\right|\right|} \tag{21}$$

3. Application of the method

The development of the methodology was motivated by an industrial case study where the crimping tools of cable manufacturing machines were tracked by an IPS.

3.1. Description of the tool-management problem

The factory that defined the research and development task produces wire harnesses for the car industry. The harnesses are composed of single wires which are attached to different terminals. The machines crimp the terminals onto the end of wires using tools (Fig. 1). Different tools are designed for different terminals. Since its establishment, the plant has manufactured a wide range of products, including 1820 different types of terminals. 2698 crimping tools have been involved in the production of these products, however, the number of tools used simultaneously over a specific period is less than one thousand.

We analysed one of the two production halls of the plant containing 23 machines and their associated storage areas. The tools may be located in these storage areas or at the machines, or being carried between them. Due to the high quantity of tools, their positions are tracked by an UWB-based IPS developed by Sunstone-RTLS Ltd.

Tool management requires information related to position data: .

- Where was each tool located and for how long was it?
- Do these locations represent a machine or a storage area?
- To what extent were the individual tools are used in the specific locations?

3.2. Data generation and transformation

The positioning system generates data that contains the horizontal and vertical coordinates ($\mathbf{x}_n = [x_{1,n}, x_{2,n}]$), a timestamp ($t_n$),and the IDs of the monitored tools. The system also provides continuous visual

information on the positions of tools according to an IPS real-time map (Fig. 2).

In our investigation, the recorded data of 15 tools between 4:50 a.m. on 01/03/2019 and 4:50 a.m. on 05/03/2019 were used. Only the data concerning the area of the smaller production hall were investigated, which included a data set of 195, 404 data points. To extract the required information from the position data, the described SFC algorithm was applied.

Supervised clustering needs training data. The training data is obtained by labeling each position according to its activity status. The activity statuses were defined as described in Section 2.2. We set Δt_{thr} to 20 s and v_{thr} to 0.1 m/s. First, data points representing moving tool (e.g., between machines or storage areas) were removed. This represented ~ 58.5% of the data or 114, 221 data points. Second, inactive states were defined. This represented ~ 8.4% of the original dataset, or 16, 502 data points. Thirdly, the active status was defined. 64, 681 data points, i.e., ~ 33.1% of the total dataset, were labeled as active.

3.3. Position data-based calculation of tool utilization

The optimal number of clusters were determined according to the methodology presented in Section 2.5. Based on the cluster validity measures depicted in Fig. 3, the optimal number of clusters was set to 22. The value of the weighting exponent m and termination tolerance ϵ were set at 2 and 10^{-3}, respectively. Fig. 4 illustrates the results of clustering the data. The position data of different tools are represented by lines in different styles and colors. The resultant cluster centers are represented by black stars, while the zones with the equal probabilities of occurrence are represented by the contour plots. As can be seen, it is possible that the different clusters can be very close to each other and their recorded position data can easily overlap due to inaccuracies in their measurement.

The activity levels of the clusters were obtained as the probability of finding active data point in that cluster (Eq. (16)). The location of machines and storage areas was identified based on these probabilities, while 0.15 was chosen as the value of threshold α: cluster in which this probability was higher than the chosen value of α were considered to be machine while all the others were considered to be storage areas. 7 locations were identified as machines in this way. In these cases, these probabilities directly calculate the utilization values at those locations, which are presented in Fig. 5.

The validity of the results is validated by engineering knowledge about layout and processes. From a perspective of the positioning system, the hall can be divided into zones along the horizontal axis of the system, and the zones can be divided into machine and storage areas along the vertical axis. The storage areas are “above", and the machines are “below". Therefore, it is expected that the found machine type clusters are also located below in Fig. 4, which is largely fulfilled.

The utilization of a single tool on a single machine can be calculated as the ratio of its working hours at that machine to the whole period. The utilization values of different tools in different clusters are shown on a heatmap (Fig. 6). With the help of the proposed system, the paths of the tools can be tracked, so the networks of the stations can be generated and visualized as a spaghetti diagram illustrated in Fig. 7.

4. Discussion: applicability of the results and suggestions for future research and development

The results illustrated the the method provides reliable information about the location-dependent utilisation of the tools. Knowledge about the utilization of the tools and machines can support scheduling maintenance activities, which can provide significant economic benefit as tool condition affects the production performance, product quality deterioration, and energy consumption [25,26]. Real-time tool monitoring can provide excellent support for zero defect manufacturing (ZDM) [8], which the help of advanced sensor technology and machine learning algorithms.

The extracted information can be used to support the lean methodology-based improvement of the production process, e.g., the application of the 5S (*Sort, Set in Order, Shine, Standardize, Sustain*) methodology. According to 5S, tools should be assigned to the storage area nearest to the machines with the highest utilization levels. Therefore, it is worth examining the distance between machines and tools. The proposed spaghetti diagrams can provide valuable information in this task.

The presented method could also be used to support the work of operators. The information extracted from the position data can implement the Smart Operator concept [27] within Operator 4.0. An intelligent personal assistant can provide the operator with helpful information to support his work, reducing production, learning, testing, and changeover times, reducing human error, and ultimately increasing worker efficiency [28]. When starting production on a given machine, based on the distances of the tools from that machine and their utilization level can be decided which tool is best to use during production. This information can be made readily available to the operator via mobile devices or smart glasses. In addition to tools, it is also worth measuring the utilization of human labor. In a case study [29], job rotation was shown to impact work performance positively, both at the individual and company level. By tracking the position of workers, it is possible to determine how much time they spend on each machine, and by linking this to tool position data, the extent to which they are involved in each type of production. This information can be used to develop an optimal job rotation plan. These techniques can increase the resilience of workers as it is introduced in the Operator 5.0 concept by making human workers more resilient against a range of factors affecting their work [30].

5. Conclusion

Advanced tool management requires accurate position-relevant information about the state of the tools. For Lean Six Sigma improvement projects, the KPIs of tools can be derived and the tool-related processes identified. Although IPS can facilitate position tracking, the accuracy of position data is not always sufficient for direct application. Therefore, the processing and cleaning of the position data are necessary.

The proposed time-weighted supervised fuzzy clustering can simultaneously explore the relevant tool locations, categorize their functions, and calculate the utilization of the tools and the machines.

The obtained information can be used in Digital Lean developments. Therefore, the positioning system can contribute towards tool management much more than shortening the time spent searching for tools and changeovers between them by facilitating the identification of underused tools, thereby reducing the extent to which they are hoarded. Furthermore, bottlenecks can be detected as evidence of overused tools, and the early detection of their overuse helps avoid or shorten unexpected stoppages due to tool failures by balancing the utilization or acquisition of spare tools at an appropriate time.

The proposed method opens new research directions pointing toward the development of Industry 5.0 solutions. Therefore, we believe it worth examining how the information extracted from positional data can improve product quality, process efficiency, sustainability, and support the work of the operators.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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